

LabVIEW-Controlled Smart Garden Robot

Complete 3-Person Hardware and Software Implementation Plan

Pot Plant Monitoring, AI Weed Detection, Leaf Health Checking, and Automatic Irrigation

Project Area	Electrical Measurements and Instrumentation Mini Project
Core Platform	Arduino Uno, NI DAQ, LabVIEW, Python AI, Local Website Dashboard
Main Task	Autonomous monitoring of predetermined potted plant stations
Navigation Method	Black tape marker detection using IR sensor
Group Size	3 members

Project Summary

This document gives a complete step-by-step implementation plan for a three-member group. The robot moves between predetermined potted plant stations, stops when a black marker is detected, captures a plant image, detects weed presence, checks basic leaf health, updates a website dashboard, measures soil moisture using a self-built sensor, calculates the moisture percentage in LabVIEW, and controls a water pump for a calculated duration.

Final Project Title

LabVIEW-Controlled Smart Garden Robot for Pot Plant Monitoring, AI Weed Detection, Leaf Health Checking, and Automatic Irrigation

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1 Final Project Scope

1.1 Main Idea

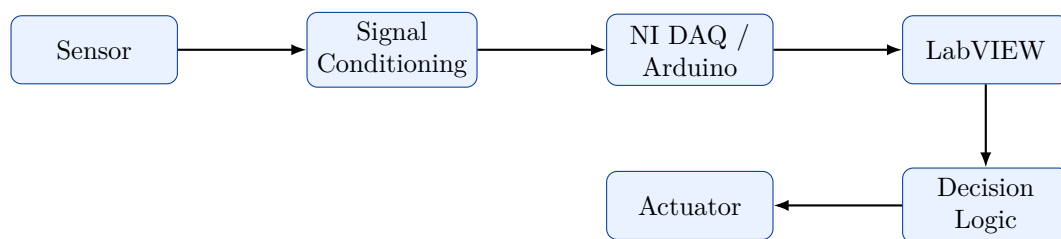
The project is designed as a controlled indoor smart garden robot for small potted plants. Instead of attempting complex outdoor autonomous navigation, the robot uses a reliable predetermined path with black tape markers placed beside each pot. At every plant station, the robot performs sensing, image analysis, decision-making, watering, data logging, and website updating.

Final System Function

Move → Stop at black marker → Capture image → Detect weed and leaf health → Measure moisture and temperature → Calculate pump duration → Water if needed → Update LabVIEW and website

1.2 Why This Scope Is Practical

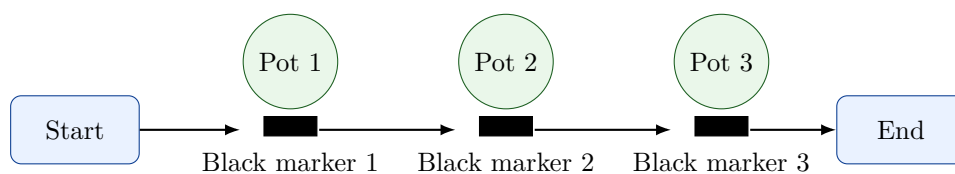
This version is suitable for a semester mini project because it is controlled, testable, and clearly connected to instrumentation. The strongest technical contribution is not only the robot movement, but the full measurement chain:



2 Final Demonstration Environment

2.1 Physical Layout

The robot will be demonstrated using three potted plants placed in a straight line. A black tape marker is placed near each pot. The IR sensor detects the marker and stops the robot at the correct measurement position.



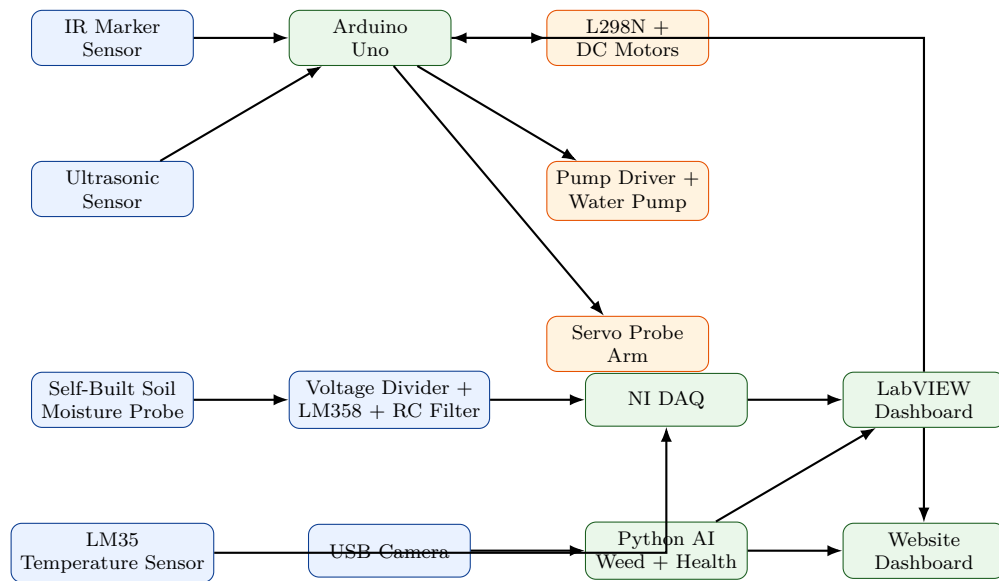
2.2 Final Demonstration Sequence

1. Start LabVIEW, Arduino serial communication, Python AI program, and website dashboard.
2. Robot moves forward from the start point.
3. IR sensor detects the first black marker and stops near Pot 1.
4. Camera captures an image of the plant and soil surface.
5. AI checks weed presence and basic leaf health condition.
6. Soil moisture probe measures the soil condition.

7. NI DAQ sends conditioned moisture and temperature signals to LabVIEW.
8. LabVIEW calculates moisture percentage and pump duration.
9. If moisture is below the target value, the pump waters the plant.
10. Website dashboard updates pot status.
11. LabVIEW logs data and sends the robot to the next station.

3 Complete System Architecture

3.1 Overall Block Diagram



3.2 Main Sensors and Actuators

Table 1: Sensors used in the final system

Sensor	Type	Purpose
Self-built soil moisture sensor	Homemade analog sensor	Measures soil moisture level using two probes, signal conditioning, and NI DAQ input.
LM35 temperature sensor	Analog sensor	Measures environmental temperature near the plant.
USB camera	Image sensor	Captures plant image for weed detection and leaf health checking.
IR marker sensor	Digital sensor	Detects black tape markers for predetermined stopping positions.
Ultrasonic sensor	Digital sensor	Optional obstacle or distance detection near the pot.

Table 2: Actuators used in the final system

Actuator	Driver	Purpose
DC motors	L298N motor driver	Move the robot between pot stations.
Mini DC water pump	MOSFET or relay driver	Supplies water according to calculated pump duration.
Servo motor	Arduino PWM	Lowers and lifts the soil moisture probe.
Buzzer and LEDs	Arduino digital pins	Shows weed alert, low moisture alert, and system status.
LED/grow light	Programmable DC supply / LabVIEW	Demonstrates light intensity control in three levels.

4 Balanced Three-Person Work Division

4.1 Division Strategy

The group already has one member working on the robot base and one member working on the AI model. To balance the workload, each member must also take additional integration tasks. The project should not be divided only as robot, AI, and sensors. Instead, each person should own one main area and support integration with the other two areas.

Important Workload Rule

Each member must produce both a working subsystem and evidence for the final report. Evidence includes circuit photos, screenshots, test tables, calibration results, code screenshots, and demonstration results.

4.2 Person 1: Robot Base, Movement, Actuator Hardware, and Arduino Control

Main Responsibility

Person 1 builds the physical robot and makes sure it can move, stop at black markers, operate the pump, move the servo probe arm, and respond to LabVIEW commands.

Table 3: Person 1 task plan

Area	Detailed Tasks
Robot chassis	Assemble 4-wheel or 2-wheel chassis, mount motors, wheels, battery, and top platform.
Motor driver	Connect DC motors to L298N and connect L298N control pins to Arduino.
Black marker stopping	Mount downward-facing IR sensor and tune it to detect black tape markers.
Station logic	Program robot to stop at Pot 1, Pot 2, and Pot 3.
Pump hardware	Mount water tank, pump, tubing, nozzle, and MOSFET/relay driver.

Servo mechanism	Build servo-lowered soil moisture probe holder.
Arduino control	Implement serial commands such as <code>MOVE</code> , <code>STOP</code> , <code>NEXT</code> , <code>PUMP_4</code> , <code>SERVO_DOWN</code> , and <code>SERVO_UP</code> .
Testing evidence	Record movement test, stop accuracy, pump test, and servo movement test.

4.3 Person 2: Camera, AI Weed Detection, Leaf Health Checking, Website, and Image Data

Main Responsibility

Person 2 develops the image capture system, weed detection, basic leaf health checking, and website dashboard. This balances the AI task by adding web display and dataset management.

Table 4: Person 2 task plan

Area	Detailed Tasks
Camera mounting	Work with Person 1 to mount camera on front mast at approximately 30–45 degrees.
Image dataset	Capture images under fixed lighting and fixed camera angle for plant-only, plant-with-weed, and soil/invalid classes.
Weed detection	Train a simple classifier or implement an OpenCV green-region detection method.
Leaf health checking	Estimate leaf condition using green, yellow, and brown color regions.
Website dashboard	Build a local Flask dashboard showing pot status, weed result, leaf health, moisture, temperature, pump duration, and image.
Data communication	Save AI and health results into CSV/JSON file for LabVIEW and website.
Testing evidence	Prepare confusion table, sample predictions, website screenshots, and image dataset examples.

4.4 Person 3: Self-Built Sensor, Signal Conditioning, NI DAQ, LabVIEW, and Control Logic

Main Responsibility

Person 3 handles the core instrumentation section: soil moisture sensor construction, signal conditioning, NI DAQ measurement, LabVIEW dashboard, calibration, pump duration algorithm, and data logging.

Table 5: Person 3 task plan

Area	Detailed Tasks
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Moisture sensor	Build two-probe stainless-steel soil moisture sensor.
Signal conditioning	Design voltage divider, LM358 buffer, and RC low-pass filter.
NI DAQ reading	Connect conditioned moisture output to AI0 and LM35 output to AI1.
Calibration	Perform dry, slightly moist, normal, and wet soil calibration.
Moisture percentage	Convert measured voltage into calibrated moisture percentage.
Temperature reading	Convert LM35 voltage into temperature in degrees Celsius.
Pump duration logic	Calculate water pump duration based on moisture deficit.
LabVIEW dashboard	Display all sensor readings, AI status, pump status, graphs, and data logging.
Testing evidence	Prepare calibration graph, repeatability test, temperature comparison, and LabVIEW screenshots.

4.5 Final Balanced Responsibility Matrix

Table 6: Balanced responsibility matrix

Task	Person 1	Person 2	Person 3
Robot chassis and motor control	Main	Support camera mounting	Support sensor placement
Black marker stop logic	Main	–	Log pot number
Pump hardware	Main	–	Pump timing calculation
Servo probe arm	Main	–	Moisture measurement timing
Camera and image capture	Mount support	Main	Display result in LabVIEW
Weed detection	–	Main	Read status file
Leaf health checking	–	Main	Display status
Website dashboard	Provide robot status	Main	Provide sensor data
Self-built moisture sensor	Mount support	–	Main
Signal conditioning	Wiring support	–	Main
NI DAQ interface	–	–	Main
LabVIEW dashboard	Serial support	AI data support	Main
Final integration	Main actuator tests	Main AI/web tests	Main measurement tests

5 Hardware Implementation Guide

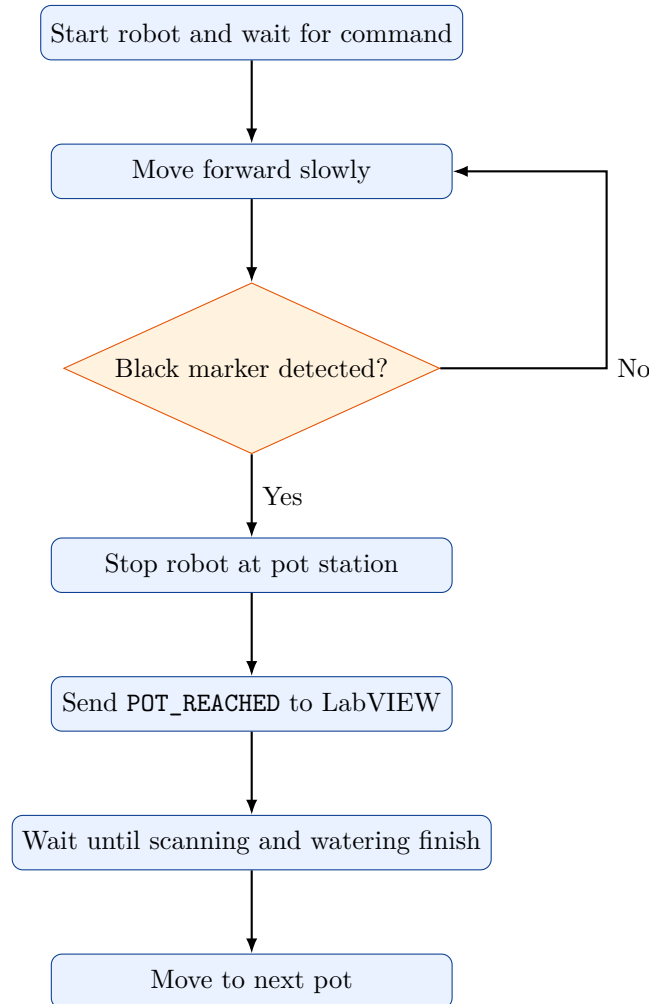
5.1 Robot Base Implementation

5.1.1 Required Components

- Robot chassis with 2 or 4 DC geared motors
- Arduino Uno
- L298N motor driver
- Battery pack for motors
- Downward-facing IR sensor
- Front-facing ultrasonic sensor
- Mini DC pump and water tank
- Relay module or MOSFET driver

- Servo motor for moisture probe arm
- Buzzer and LED indicators

5.1.2 Robot Movement Logic



5.2 Self-Built Soil Moisture Sensor

5.2.1 Sensor Principle

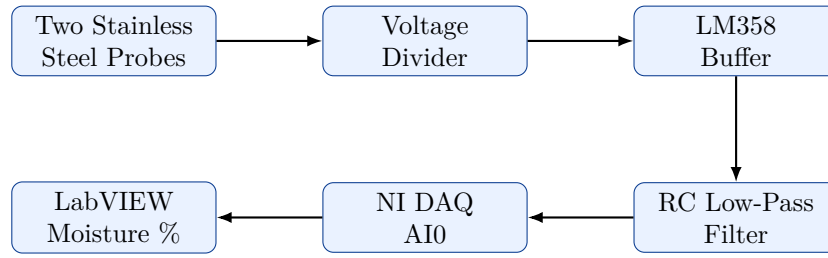
The soil moisture sensor is built using two stainless-steel probes. Soil moisture affects the electrical resistance between the probes. Dry soil has higher resistance, while wet soil has lower resistance. The probe signal is converted into a voltage using a voltage divider, buffered using an LM358 op-amp, filtered using an RC low-pass filter, and measured by the NI DAQ.

Voltage Divider Concept

$$V_{out} = V_{in} \times \frac{R_{fixed}}{R_{soil} + R_{fixed}}$$

$$R_{soil} \downarrow \Rightarrow V_{out} \uparrow$$

5.2.2 Signal Conditioning Chain



5.2.3 Moisture Calibration Equation

Use dry and wet soil reference voltages to convert sensor voltage into moisture percentage.

Relative Moisture Percentage

$$M(\%) = \frac{V_{sample} - V_{dry}}{V_{wet} - V_{dry}} \times 100$$

$$0\% \leq M(\%) \leq 100\%$$

Table 7: Recommended soil moisture calibration table

Soil Condition	Measured Voltage	Assigned Moisture
Dry soil	V_{dry}	0%
Slightly moist soil	V_1	30%
Normal moist soil	V_2	60%
Wet soil	V_{wet}	100%

5.3 Temperature Measurement Using LM35

The LM35 is selected because it gives an analog output suitable for NI DAQ measurement and LabVIEW display.

LM35 Temperature Conversion

$$T(^{\circ}C) = \frac{V_{out}}{0.01}$$

For example, if $V_{out} = 0.30\text{ V}$,

$$T = \frac{0.30}{0.01} = 30^{\circ}C$$

5.4 Automatic Water Pump Control

The pump duration is calculated from the difference between target moisture and measured moisture.

Pump Duration Algorithm

$$D = M_{target} - M_{measured}$$

If $D \leq 0$, pump is OFF.

$$D > 0 \Rightarrow t_{pump} = K \times D$$

$$t_{pump} = \min(KD, 8 \text{ s})$$

Recommended values:

$$M_{target} = 60\%, \quad K = 0.12 \text{ s}/\%$$

Table 8: Backup rule-based pump control table

Moisture Range	Soil Condition	Pump Duration
0–20%	Very dry	8 s
21–40%	Dry	5 s
41–60%	Slightly low	2 s
> 60%	Normal	0 s

6 AI Weed Detection and Leaf Health Checking

6.1 Camera Setup

The camera should be fixed to the front mast of the robot and angled downward toward the pot at approximately 30–45 degrees. The same angle and lighting should be used for training and final demonstration.

6.2 Weed Detection Classes

Table 9: Weed detection dataset classes

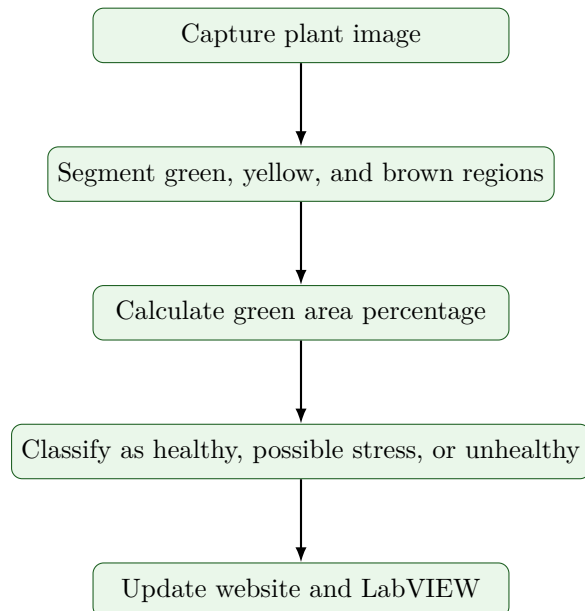
Class	Meaning	Minimum Images
Plant only	No weed detected	40–60
Plant with weed	Weed detected	40–60
Soil/invalid	No valid plant image	20–30

6.3 Leaf Health Estimation

For the mini project, leaf health checking should be kept simple and reliable. The system estimates leaf condition based on visible color regions rather than complex disease diagnosis.

Table 10: Basic leaf health classification

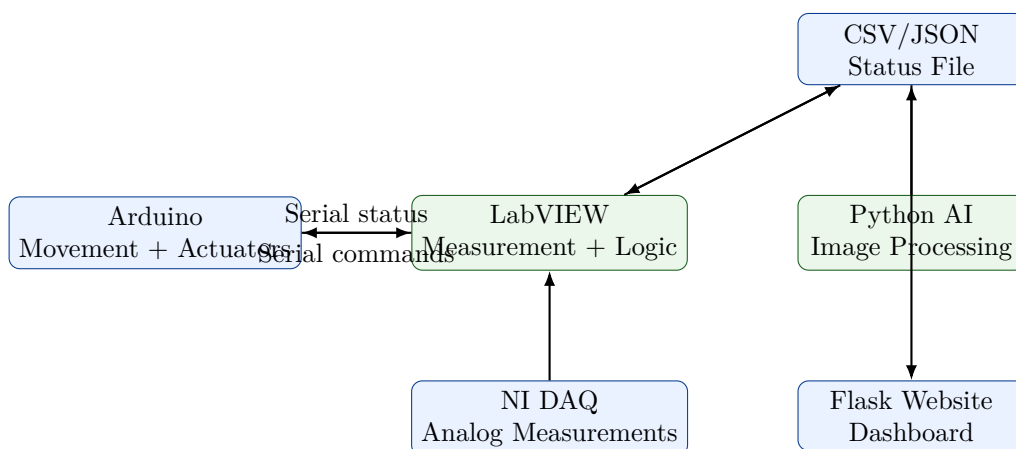
Green Leaf Area	Health Result
> 75%	Healthy
50–75%	Possible stress
< 50%	Unhealthy



7 LabVIEW, Website, and Communication Plan

7.1 Recommended Communication Architecture

A simple and reliable architecture should be used. Arduino handles movement and actuators, LabVIEW handles measurements and control logic, Python handles image processing, and the website displays the latest pot status.



7.2 LabVIEW Dashboard Items

- Current pot number

- Robot status: moving, stopped, scanning, watering, completed
- Soil moisture voltage
- Soil moisture percentage
- Temperature in degrees Celsius
- Weed status
- Leaf health status
- Pump duration
- Pump ON/OFF indicator
- Moisture vs. time graph
- Temperature vs. time graph
- Manual controls: start, stop, next pot, pump test, emergency stop

7.3 Website Dashboard Format

Table 11: Website dashboard table format

Pot	Weed Status	Leaf Health	Moisture	Temperature	Pump
Pot 1	Weed detected	Healthy	35%	30.2°C	4 s
Pot 2	No weed	Possible stress	68%	29.5°C	0 s
Pot 3	Weed detected	Unhealthy	42%	31.0°C	2 s

7.4 Arduino Command List

Table 12: Commands sent from LabVIEW to Arduino

Command	Meaning
MOVE	Move robot forward
STOP	Stop robot immediately
NEXT	Move to the next pot station
SERVO_DOWN	Lower soil moisture probe
SERVO_UP	Lift soil moisture probe
PUMP_2	Turn pump ON for 2 seconds
PUMP_4	Turn pump ON for 4 seconds
PUMP_8	Turn pump ON for 8 seconds
BUZZER_ON	Turn buzzer ON
BUZZER_OFF	Turn buzzer OFF

Table 13: Messages sent from Arduino to LabVIEW

Message	Meaning
POT_1_REACHED	Robot reached Pot 1
POT_2_REACHED	Robot reached Pot 2
POT_3_REACHED	Robot reached Pot 3
PUMP_DONE	Watering process finished
PROBE_DOWN_DONE	Probe inserted into soil
PROBE_UP_DONE	Probe lifted from soil
OBSTACLE_DETECTED	Obstacle detected and robot stopped

8 Final Step-by-Step Implementation Plan

8.1 Step 1: Fix the Demo Environment

1. Place three potted plants in a straight line.
2. Place one black tape marker beside each pot.
3. Mark the robot start point and end point.
4. Fix lighting for camera capture.
5. Keep laptop, NI DAQ, LabVIEW, and website ready near the robot path.

8.2 Step 2: Build Robot Movement System

1. Assemble robot chassis, motors, wheels, and battery.
2. Connect motors to L298N motor driver.
3. Connect L298N to Arduino Uno.
4. Test forward movement and stop function.
5. Add downward-facing IR sensor.
6. Test black marker detection.
7. Program robot to stop at three markers.

8.3 Step 3: Build Moisture Sensor and Conditioning Circuit

1. Make two stainless-steel probes.
2. Build voltage divider circuit.
3. Add LM358 voltage follower.
4. Add RC low-pass filter.
5. Connect output to NI DAQ AI0.
6. Test output voltage with dry and wet soil.

8.4 Step 4: Calibrate Soil Moisture Sensor

1. Prepare dry, slightly moist, normal moist, and wet soil samples.
2. Record output voltage for each condition.
3. Assign moisture percentages to each condition.
4. Use the calibration equation in LabVIEW.
5. Verify that output stays between 0% and 100%.

8.5 Step 5: Add Temperature Measurement

1. Connect LM35 output to NI DAQ AI1.
2. Convert voltage into temperature using LabVIEW.
3. Compare with a normal thermometer.
4. Display temperature on the dashboard.

8.6 Step 6: Add Pump and Irrigation System

1. Mount water tank at the rear of the robot.
2. Fix mini DC pump near the tank.
3. Connect tube from pump to front nozzle.
4. Build MOSFET or relay driver circuit.
5. Add flyback diode for protection.
6. Test pump manually.
7. Test calculated pump duration from LabVIEW.

8.7 Step 7: Add Camera and Image Processing

1. Mount camera at front-top of robot.
2. Keep camera angle fixed.
3. Capture images for each pot.
4. Train or program weed detection.
5. Program leaf health checking.
6. Save output results to CSV/JSON.

8.8 Step 8: Build Website Dashboard

1. Create Flask dashboard.
2. Display pot ID, image, weed status, leaf health, moisture, temperature, and pump duration.
3. Read data from CSV/JSON file.
4. Update website after each pot scan.

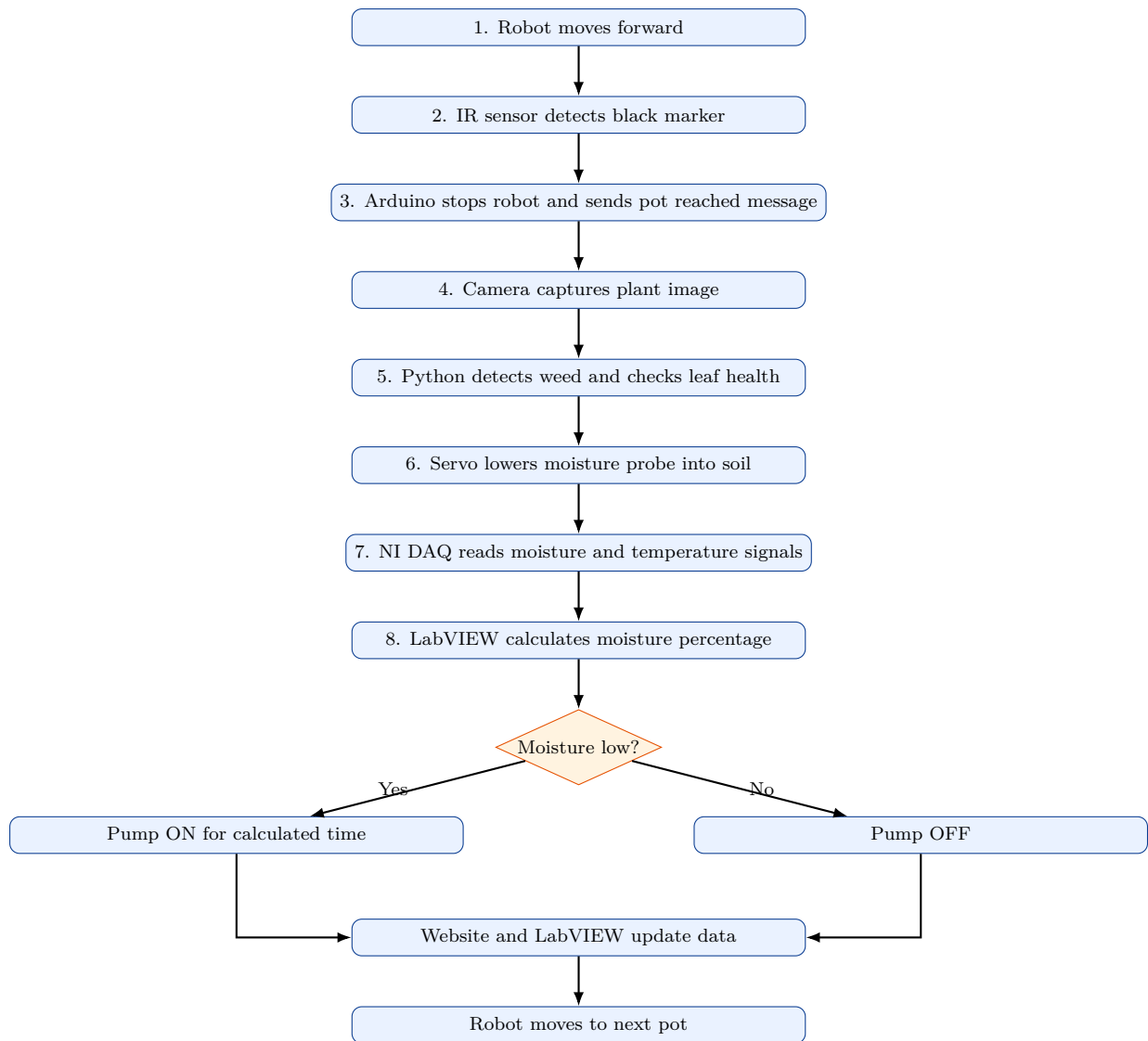
8.9 Step 9: Build LabVIEW Dashboard

1. Read moisture voltage from NI DAQ AI0.
2. Read temperature voltage from NI DAQ AI1.
3. Convert sensor voltage into engineering values.
4. Read AI result from CSV/JSON.
5. Calculate pump duration.
6. Send pump and robot commands to Arduino.
7. Log all data into CSV file.

8.10 Step 10: Full System Integration

1. Test one complete pot cycle first.
2. After Pot 1 works, expand to Pot 2 and Pot 3.
3. Check that robot stops correctly at every marker.
4. Check that image, sensor values, LabVIEW, pump, and website update in the correct order.
5. Record final demonstration video.

9 One-Pot Final Control Sequence



10 Seven-Day Group Implementation Schedule

Table 14: Seven-day implementation schedule

Day	Person 1	Person 2	Person 3
Day 1	Assemble robot chassis, motors, L298N, and Arduino. Test basic movement.	Test camera, create dataset folders, capture first plant images.	Build moisture probe and voltage divider. Test voltage change.
Day 2	Add IR sensor and black marker stop logic.	Continue image dataset and start leaf health color method.	Add LM358 buffer and RC filter. Connect to NI DAQ.
Day 3	Mount pump, tank, tubing, and pump driver.	Train first weed detection model or implement OpenCV method. Start Flask website.	Calibrate moisture sensor and add LM35 temperature reading.

Day 4	Add servo probe arm and pump duration test.	Connect AI output to CSV/JSON and website.	Build LabVIEW dashboard and pump duration calculation.
Day 5	Test Pot 1 to Pot 3 movement and improve stopping accuracy.	Website displays all three pot results. Test weed and health result.	LabVIEW displays sensor data, AI result, and logging.
Day 6	Test movement reliability, pump, servo, and actuator safety.	Test AI accuracy, website update speed, and image saving.	Test moisture repeatability, temperature accuracy, and control logic.
Day 7	Clean wiring and prepare robot demo.	Prepare AI/website screenshots and explanation.	Prepare LabVIEW screenshots, calibration graph, and sensor explanation.

11 Testing and Performance Evaluation

11.1 Robot Testing

Table 15: Robot performance testing

Test	Expected Result
Forward movement test	Robot moves smoothly without excessive vibration.
Black marker detection	Robot stops when IR sensor detects marker.
Stopping accuracy	Robot stops close enough for camera and probe alignment.
Pump test	Pump turns ON and OFF according to command.
Servo test	Moisture probe enters and exits soil correctly.
Cycle time test	Robot completes each pot station within acceptable time.

11.2 Sensor Testing

Table 16: Sensor performance testing

Test	Expected Result
Dry soil test	Sensor gives stable dry reference voltage.
Wet soil test	Sensor gives clear voltage difference compared with dry soil.
Moisture percentage test	LabVIEW output remains between 0% and 100%.
Repeatability test	Same soil sample gives similar readings in repeated trials.
Temperature comparison	LM35 reading is close to thermometer reading.
Response time	Reading stabilizes within a short measurement period.

11.3 AI and Website Testing

11.4 Control Testing

Table 17: AI and website testing

Test	Expected Result
Plant-only image	Output should be no weed.
Plant with weed image	Output should be weed detected.
Healthy green leaf image	Output should be healthy.
Yellow/brown leaf image	Output should be possible stress or unhealthy.
Website update test	Correct pot row updates after each scan.
Image display test	Latest pot image is visible on website.

Table 18: Pump control testing

Test	Expected Result
Moisture below target	Pump turns ON for calculated duration.
Moisture above target	Pump remains OFF.
Maximum duration limit	Pump duration does not exceed 8 seconds.
Watering effect	Moisture percentage increases after watering.
Manual override	User can manually turn pump ON/OFF from LabVIEW.

12 Final Data Logging Format

The system should save one row for each pot scan.

Table 19: Final CSV logging format

Time	Pot ID	Moisture	Temp.	Weed	Leaf Health	Pump	Action
10:01	Pot 1	35%	30.2°C	Weed	Healthy	4 s	Pump ON
10:05	Pot 2	68%	29.5°C	No weed	Possible stress	0 s	No action
10:09	Pot 3	42%	31.0°C	Weed	Unhealthy	2 s	Pump ON

13 Priority Plan Under Time Constraint

13.1 Must Complete

1. Robot moves and stops at black markers.
2. Camera captures image at each pot.
3. AI gives weed/no weed result.
4. Website updates pot result.
5. Self-built soil moisture sensor gives calibrated output.
6. LabVIEW calculates moisture percentage.
7. Pump watering duration is calculated and controlled.
8. Temperature reading is shown in LabVIEW.
9. Final data is logged.

13.2 Should Complete

1. Leaf health checking using color-based analysis.
2. Servo-lowered moisture probe.
3. Website image display.
4. Live graphs in LabVIEW.

13.3 Optional Only If Time Remains

1. Advanced object detection model.
2. Weed removal arm.
3. Cloud database.
4. Mobile app.
5. Full autonomous navigation without markers.

14 Final Presentation Roles

14.1 Person 1 Presentation

I developed the robot movement system, black marker stopping method, actuator wiring, pump system, and servo-based soil probe mechanism. The robot moves between predetermined pot stations and stops when the IR sensor detects a black marker.

14.2 Person 2 Presentation

I developed the camera-based weed detection and basic leaf health checking system. The system captures an image at each pot, detects weed presence, estimates leaf condition using image features, and updates the website dashboard.

14.3 Person 3 Presentation

I developed the self-built soil moisture sensor, signal conditioning circuit, NI DAQ interface, LabVIEW dashboard, calibration equation, temperature measurement, and automatic pump duration control based on soil moisture percentage.

15 Final Group Integration Statement

Statement for Report or Presentation

Our system is implemented as a controlled smart garden robot for potted plants. The robot moves between predetermined plant stations using black tape markers for reliable stopping. At each station, the system captures a plant image for weed detection and basic leaf health checking, measures soil moisture using a self-built sensor with signal conditioning, reads temperature using an LM35 sensor, calculates the soil moisture percentage in LabVIEW, and automatically controls a water pump for a calculated duration. The website dashboard updates the weed status, leaf health, moisture level, temperature, and watering action for each pot.

16 Conclusion

This implementation plan provides a balanced and practical approach for a three-member group. The project can be completed by first making one pot station work successfully, then expanding to three pot stations. The final system clearly demonstrates measurement, calibration, signal conditioning, DAQ-based acquisition, LabVIEW control, Arduino actuation, AI-assisted plant monitoring, website updating, and automatic irrigation. This makes the project suitable for an Electrical Measurements and Instrumentation mini project.